Final Exam

History of IPv6

1. What organization has ultimate responsibility for the direction of the Internet?
    a. IAB
    b. IETF
    c. ISOC
    d. ISO

2. In the early 1990s, which two protocol suites were in competition to be the protocol suite of the Internet?
    a. DecNet
    b. TCP/IP
    c. OSI
    d. ISO

3. In 1992, which layer 3 protocol was proposed by the IAB to replace IPv4?
    a. TCP/IP's IPv5
    b. TCP/IP's IPv6
    c. OSI's CLNP
    d. None, they would request proposals

4. What was IETF's fear concerning the next generation protocol not being IP?
    a. They would have to learn something new.
    b. They would lose change control over the protocol and be subjected to ISO's complex standards process.
    c. They would be out of a job.
    d. The TCP/IP protocol suite including all upper layer protocols would be replaced by OSI protocols.

8. What name did the IANA announce for the new IPng protocol?
    a. IPv5
    b. IPv6
    c. IPv4.1
    d. IPv7

Comparing IPv4 and IPv6

1. Which of the following IPv4 fields are similar to those in the IPv6 header?
    a. version
    b. IHL
    c. Types of Service (DSCP)
    d. Total Length
    e. Identification
    f. Flag
    g. Offset
    h. TTL
    i. Protocol
    j. Header Checksum
    k. Source Address
    l. Destination address
    m. Options
    n. Padding

2. Which protocol has a fixed header of 40 bytes?
    a. IPv4
    b. IPv6

3. When referring to the payload length, which IP protocol only includes the number of bytes in the data (payload), and not the main IP header?
    a. IPv4
    b. IPv6

4. Which protocol allows routers to fragment packets?
    a. IPv4
    b. IPv6

5. Which protocol provides a checksum to verify that the header was not modified in transit?
    a. IPv4
    b. IPv6

6. Which of the following can be used as a source address in IPv4 and IPv6? (Choose all that apply)

a. unicast
b. multicast
c. broadcast
d. anycast

7. Which of the following can be used as a destination address in IPv6? (Choose all that apply)

a. unicast
b. multicast
c. broadcast
d. anycast

Global Unicast Addresses

1. Global Unicast Addresses (GUA): (Choose all that apply)

a. Cannot be routed off the link
b. Are globally unique
c. Are globally routable
d. Are similar to private IPv4 addresses
e. Are similar to public IPv4 addresses
f. Are not installed in a router's routing table

2. The first hexadecimal digit of a GUA is a: (Choose all that apply)

a. 1
b. 2
c. 3
d. Any hexadecimal digit
e. Any hexadecimal digit other than F

3. Which GUA is reserved for documentation?

2001:db8::/32

4. Match the IPv6 term with its IPv4 equivalent:

IPv6:

a. Prefix

b. Prefix length

c. Interface ID

Host portion of the address

Network portion of the address

Subnet mask

5. What part of the GUA is allocated by the ISP or provider of the IPv6 prefix?

Global Routing Prefix

6. What part of the GUA is between the Global Routing Prefix and the Interface ID?

Subnet ID

7. How is does the Subnet ID field differ from subnetting in IPv4?

You do not have to borrow bits from the Interface ID (host portion) to create subnets.

8. **2001:0db8:bee5::/48 subnetted using /52**

How many subnets does this create? 16 subnets

```
2001:0DB8:BEE5:0000::/52
2001:0DB8:BEE5:1000::/52
2001:0DB8:BEE5:2000::/52
2001:0DB8:BEE5:3000::/52
2001:0DB8:BEE5:4000::/52
2001:0DB8:BEE5:5000::/52
2001:0DB8:BEE5:6000::/52
2001:0DB8:BEE5:7000::/52
2001:0DB8:BEE5:8000::/52
2001:0DB8:BEE5:9000::/52
2001:0DB8:BEE5:A000::/52
2001:0DB8:BEE5:B000::/52
2001:0DB8:BEE5:C000::/52
2001:0DB8:BEE5:D000::/52
2001:0DB8:BEE5:E000::/52
2001:0DB8:BEE5:F000::/52
```

9. **2001:0db8:bee5::/48 subnetted using /56**

How many subnets does this create? 256 subnets

10. **2001:0db8:bee5::/48 subnetted using /64**

How many subnets does this create? 65,536 subnets

10. Why are /64 Interface IDs recommended in most LANs?
SLAAC can be used which requires a /64, using EUI-64 or randomly generated 64-bit Interface ID.

11. Describe steps in the EUI-64 process.
1. Divide MAC Address in half between OUI and Device ID
2. Insert FF-FE
3. Flip the 7th bit

12. List the ways a device can be configured with a GUA.
1. Static (Cisco routers also allow static with EUI-64)
2. SLAAC (EUI-64 and random 64-bit Interface ID)
3. Stateful DHCPv6

Link-Local Unicast Addresses

1. Link-local addresses: (Choose all that apply)
    a. Cannot be routed off the link
    b. Are globally unique
    c. Are globally routable
    d. Are Similar to private IPv4 addresses

e. Are similar to public IPv4 addresses

f. Are not installed in a router's routing table

2. What is the range of the first hextet for a link-local address?

fe80 through febf

3. A link-local address can be a source IPv6 address. (T/F)

a. True

b. False

3. A link-local address can be a destination IPv6 address. (T/F)

a. True

b. False

4. Link-local addresses are optional on IPv6 devices. (T/F)

a. True

b. False

5. Which of the following are uses of a link-local address?

a. Default gateway address when obtained from a Router Advertisement message

b. As the source address in dynamic routing protocol messages

c. As the source address in a Router Advertisement message

d. As the source address in a Router Solicitation message

6. Why do some operating systems require the use of an exit interface when pinging a link-local address?

Because the link-local address can be out any interface. They only have to be unique on the link.

7. Windows, Linux, and Mac OS clients IPv6 enabled by default. What tells you this?

They have an IPv6 link-local address.

8. How does a device automatically create its own IPv6 link-local address?

fe80:: + 64-bit Interface ID (EUI-64 or randomly generated 64-bit value)

9. Can the same link-local address be configured on more than one interface? Why or why not.

Yes. The link-local address only has to be unique on the link.

10. Which devices should have a statically configured link-local address to help recognize and remember the address?

Routers

11. The Cisco IOS **ipv6 enable** command is required to configure any IPv6 address on the router's interface. (T/F)

a. True

b. False

12. What is the purpose of the Cisco IOS **ipv6 enable** command?

Allows the router to automatically configure a link-local address using EUI-64 without having a global unicast address.

13. When a Cisco router's interface obtains a global unicast address (statically or dynamically), it will automatically configure a link-local address on that interface. (T/F)

a. True

b. False

Multicast Addresses

1. What are the two types of IPv6 multicast addresses?

   Assigned and solicited-node

2. IPv4 224.0.0.0/4 address are equivalent to which type of IPv6 multicast address:

   Assigned

3. What is the range of the first two hexadecimal digits of a multicast address?

   ff00::/8

4. What does the third digit in the multicast address represent?

   The flag. 0 is a permanent, well-known multicast address assigned by IANA. Includes both assigned and solicited-node multicast addresses. 1 is non-permanently-assigned, “dynamically" assigned multicast address.

5. What is does the fourth digit in the multicast address represent?

   Scope (partial list):

   - 0 Reserved
   - 1 Interface-Local scope
   - 2 Link-Local scope
   - 5 Site-Local scope
   - 8 Organization-Local scope
   - E Global scope

6. What is the multicast address for all IPv6 devices? What would make a device be an IPv6 device?

   ff02::1 When the device has at least an IPv6 link-local address

7. What is the multicast address for all IPv6 routers? What would make a device be an IPv6 router?

ff02::2 When a router (Cisco IOS) is configured with the ipv6 unicast-routing command.

8. What does the "2" represent in the ff02:: multicast addresses?

Link-local scope, meaning not to be routed off the link.

9. Which addresses generate an associated solicited-node multicast address? (Choose all that apply)

a. global unicast address
b. link-local address
c. anycast address
d. assigned multicast address

10. A solicited-node multicast address uses the prefix ff02:0:0:0:0:1ff00::/104. How are the lower-end 24 bits determined?

From the low-order 24 bits of the unicast or anycast address.

11. Given a solicited-node multicast address of ff02::1:ff11:2233, what would be the associated Ethernet MAC address?

33-33-ff-11-22-33

12. What is the advantage of using a solicited-node multicast address instead of a broadcast address?

The Ethernet NIC card can filter, accept, only frames with destination MAC addresses that are map to a solicited-node multicast address associated with one of its unicast addresses.

13. Is it possible for two devices on the same network to have the same solicited-node multicast address?

Yes, if the low-order 24 bits of their unicast addresses are the same. This is not a problem because the data (ICMPv6) will contain the target unicast address.

14. What do Ethernet switches do when a frame has a destination MAC address that is a solicited-node multicast address?

Floods it out all ports except the incoming port.

15. When are solicited-node multicast addresses used?

ICMPv6 neighbor discovery, similar to ARP in IPv4

16. A solicited-node multicast address uses the prefix ff02:0:0:0:0:1ff00::/104. What does the 2 in the first hextet indicate?

The address has link-local scope and is not to be routed off the link.

IPv6 Addressing

```
C:\Users\Win7Admin>ipconfig

Windows IP Configuration


Ethernet adapter Local Area Connection:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2001:db8:cafe:1:250:56ff:feaf:c90e
   Link-local IPv6 Address . . . . . : fe80::250:56ff:feaf:c90e%11
   Autoconfiguration IPv4 Address. . : 169.254.25.150
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : fe80::1%11
```

1. Using the exhibit above, what process did the Windows host most likely use to obtain its IPv6 address?
   a. Static
   b. SLAAC using EUI-64
   c. SLAAC using the privacy extension
   d. Stateful DHCPv6

```
C:\Users\Win7Admin>ipconfig

Windows IP Configuration


Ethernet adapter Local Area Connection:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2001:db8:cafe:1:7492:e5a9:2bbd:1996
   Link-local IPv6 Address . . . . . : fe80::7492:e5a9:2bbd:1996%11
   Autoconfiguration IPv4 Address. . : 169.254.25.150
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : fe80::1%11
```

2. Using the exhibit above, what process did the Windows host most likely use to obtain its IPv6 address? A couple of possibilities…
   a. Static
   b. SLAAC using EUI-64
   c. SLAAC using the privacy extension
   d. Stateful DHCPv6

3. Router Advertisements include several flags. Which of the following flags are on by default in a Cisco RA message? (choose all that applies)
   a. Autonomous address configuration flag
   b. Other configuration flag
   c. Managed configuration flag
   d. On-Link flag

4. Which RA flag is set to suggest to a receiving device that it should use SLAAC?
   a. Autonomous flag

b. Other configuration flag
c. Managed configuration flag
d. On-Link flag

5. Which RA flagis set to suggest to a receiving device to use **stateless DHCPv6** to obtain information such as a DNS server address?

a. Autonomous address configuration flag
b. Other configuration flag
c. Managed configuration flag
d. On-Link flag

6. Which RA flag is set to suggest to a receiving device to use **stateful DHCPv6** to obtain addressing information?

a. Autonomous address configuration flag
b. Other configuration flag
c. Managed configuration flag
d. On-Link flag

7. Which of the following information can be obtained from a Router Advertisement? (choose all that applies)

a. Default gateway
b. Prefix
c. Prefix-length
d. MAC address
e. DNS addresses
f. Domain names

16. What information **can only be obtained dynamically** from an RA message and not from a stateful DHCPv6 server?

a. Default gateway
b. Prefix
c. Prefix-length
d. MAC address
e. DNS addresses
f. Domain names

17. Which of the following statements is correct regarding a device dynamically receiving a **default gateway address**? (choose all that applies)

a. It is a link-local address of a RA message
b. It is a global unicast address of a RA message
c. It is the source IPv6 address of the RA message
d. It is contained within the ICMPv6 message
e. It can obtained from a stateless DHCPv6 server
f. It can obtained from a stateful DHCPv6 server
g. It can only be obtained from a RA message

18. When a device is configured with a statically IPv6 address, it cannot have a dynamically obtained IPv6 address?

a. True
b. False

19. How can a device ensure that an IPv6 address is unique?

a. Using Duplicate Address Detection, but only on dynamically obtained addresses
b. Using Duplicate Address Detection, but only on statically configured addresses
c. Using Duplicate Address Detection, on any global unicast address
d. Using Duplicate Address Detection, on any unicast address

ICMPv6

1. When an IPv6 router receives a packet and the MTU of the outgoing link is smaller than the size of the packet, the router will:
   a. Drop the packet, no messages sent
   b. Drop the packet and send an ICMPv6 Packet Too Big message back to the source
   c. Fragment the packet and send an ICMPv6 Packet Too Big message back to the source
   d. Fragment the packet and send an ICMPv6 Packet Too Big message to the destination

2. What information is included in the ICMPv6 Packet Too Big message?
   a. MTU of the egress link needed to forward the packet
   b. MTU of the ingress link the packet was received
   c. MTU of the ingress link of the final destination
   d. The largest MTU of any link between the router and the final destination

3. Which ICMPv6 message is sent when a router drops a packet because the IPv6 hop limit was decremented to 0?
   a. Destination unreachable
   b. Packet Too Big
   c. Time Exceeded
   d. Parameter Problem

ICMPv6 ND

1. Router Solicitation, Router Advertisement, Neighbor Solicitation, Neighbor Advertisement, and Redirect messages are all part of:
   a. ICMPv6 Neighbor Discovery
   b. ICMPv6 Duplicate Address Detections
   c. ICMPv6 Dynamic Address Allocation
   d. ICMPv6 Address Allocation

2. Router Solicitation and Router Advertisement messages are for:
   a. Router-to-device messaging
   b. Device-to-device messaging

3. Neighbor Solicitation and Neighbor Advertisement messages are for:
   a. Router-to-device messaging
   b. Device-to-device messaging

4. When a host knows the IPv6 address but needs to know the MAC address, it will send out a:
   a. Neighbor Solicitation message to all-IPv6 multicast address
   b. Neighbor Solicitation message to the solicited-node multicast address
   c. Neighbor Advertisement message to all-IPv6 multicast address
   d. Neighbor Advertisement message to the solicited-node multicast address

5. When a device responds to a request for its MAC address, it will send out a:
   a. Neighbor Solicitation message to the solicited-node multicast address of the device
   b. Neighbor Solicitation message to the unicast address of the device
   c. Neighbor Advertisement message to the solicited-node multicast address of the device
   d. Neighbor Advertisement message to the unicast address of the device

6. Why would multiple devices process the same Neighbor Solicitation message?
   a. The destination IPv6 address is a broadcast
   b. The destination IPv6 address is an all-IPv6 devices multicast
   c. The destination IPv6 address is a solicited-node multicast, which is received by all IPv6 devices
   d. The destination IPv6 address is a solicited-node multicast with the last 24 bits of the address matching one of its unicast addresses

SLAAC Address States

1. When a SLAAC address can be used for new connections, what is the state of that address?
   a. Tentative
   b. Preferred
   c. Deprecated
   d. Invalid

2. When a SLAAC address cannot be used for new connections, but can be used for existing connections, what is the state of that address?
   a. Tentative
   b. Preferred
   c. Deprecated
   d. Invalid

3. When the uniqueness of a SLAAC address (or any unicast address) is being determined using DAD, what is the state of that address?
   a. Tentative
   b. Preferred
   c. Deprecated
   d. Invalid

4. When a SLAAC address that was previously used for connections can no longer be used and there are no existing connections, what is the state of that address?
   a. Tentative
   b. Preferred
   c. Deprecated
   d. Invalid

5. When is SLAAC address is considered to be in valid state? (choose all that applies)
   a. Tentative
   b. Preferred
   c. Deprecated
   d. Invalid

35. The timers, “lifetimes”, are included in the RA message? (choose two)
   a. Tentative
   b. Preferred
   c. Deprecated
   d. Invalid
   e. Valid

Neighbor Discovery – Address Resolution

1. The equivalent of an IPv4 ARP Request in IPv6 is a:
   a. Neighbor Solicitation message
   b. Neighbor Advertisement message
   c. Router Solicitation message
   d. Router Advertisement message

2. The equivalent of an IPv4 ARP Reply in IPv6 is a:
   a. Neighbor Solicitation message
   b. Neighbor Advertisement message
   c. Router Solicitation message
   d. Router Advertisement message

3. The IPv6 table/cache used to store IPv6 addresses and their corresponding MAC address is known as the:
   a. ARP table/cache
   b. MAC address table/cache
   c. Destination table/cache
   d. Neighbor table/cache

4. What is the state of the neighbor table when a Neighbor Advertisement message has been received for a corresponding Neighbor Solicitation message?
   a. Incomplete
   b. Reachable
   c. Stale
   d. Delay
   e. Probe

5. What is the state of the neighbor table when an entry marked as reachable has been in the table for over 30 seconds without the device sending a frame to this destination?
   a. Incomplete
   b. Reachable
   c. Stale
   d. Delay
   e. Probe

6. What Cisco IOS command displays the IPv6 neighbor table?
   a. show ipv6 arp
   b. show ipv6 neighbors
   c. show ipv6 neighbor-table
   d. show neighbors

7. What Windows **netsh interface ipv6** command displays the IPv6 neighbor table?
   a. **netsh interface ipv6** show ipv6 arp
   b. **netsh interface ipv6** show ipv6 neighbors
   c. **netsh interface ipv6** show ipv6 neighbor-table
   d. **netsh interface ipv6** show neighbors

8. What debug command is used to view Router Solicitations, Router Advertisements, Neighbor Solicitations and Neighbor Advertisement messages?
   a. debug ipv6 packet
   b. debug ipv6 icmp
   c. debug ipv6 icmpv6
   d. debug ipv6 nd